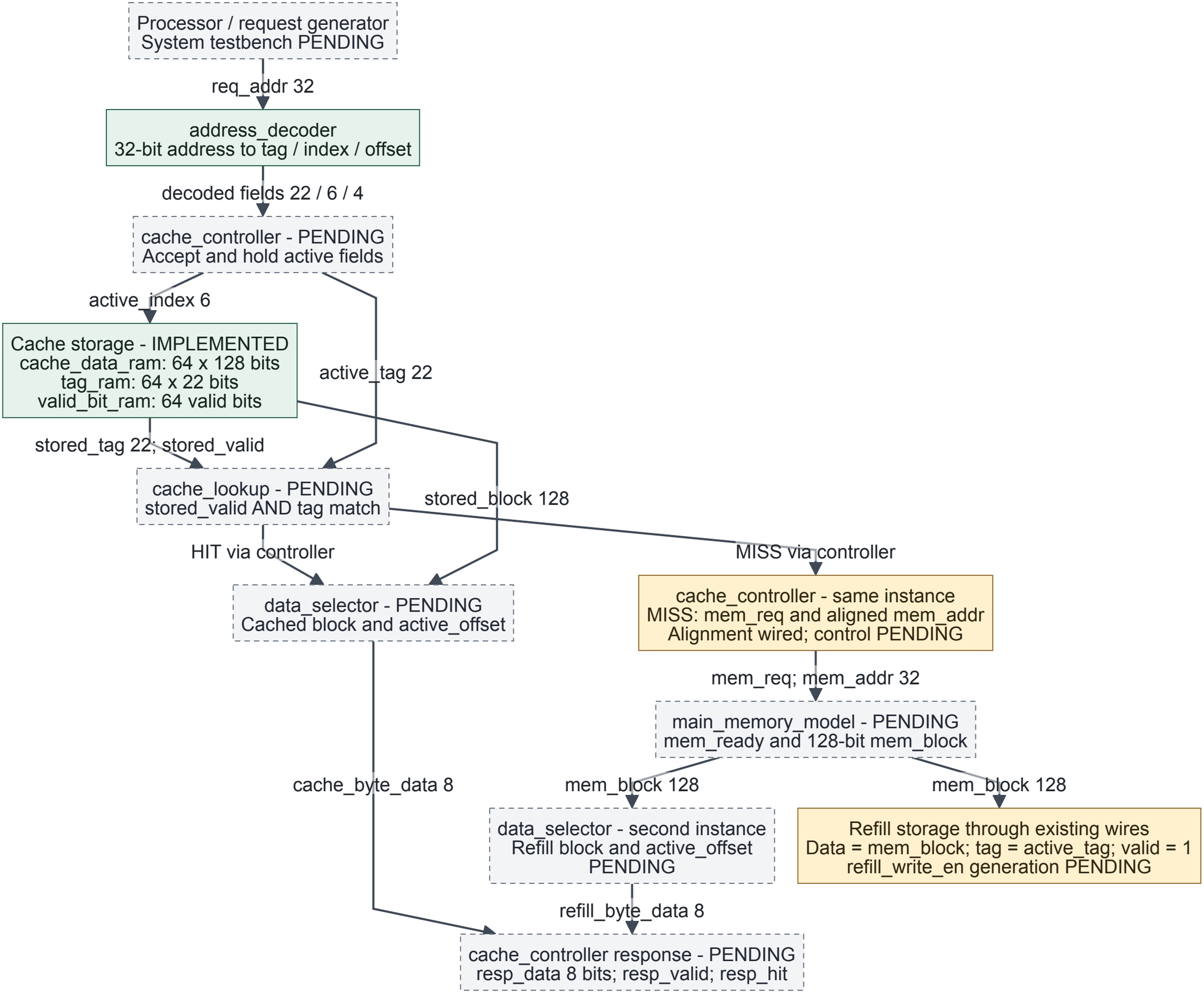
Processor / request generator
System testbench PENDING
req_addr 32
address_decoder
32-bit address to tag / index / offset
decoded fields 22 / 6 / 4
cache_controller - PENDING
Accept and hold active fields
active_index 6
Cache storage - IMPLEMENTED
cache_data_ram: 64 x 128 bits
tag_ram: 64 x 22 bits
valid_bit_ram: 64 valid bits
active_tag 22
stored_tag 22; stored_valid
cache_lookup - PENDING
stored_valid AND tag match
stored_block 128
HIT via controller
MISS via controller
data_selector - PENDING
Cached block and active_offset
cache_controller - same instance
MISS: mem_req and aligned mem_addr
Alignment wired; control PENDING
mem_req; mem_addr 32
main_memory_model - PENDING
mem_ready and 128-bit mem_block
cache_byte_data 8
mem_block 128
mem_block 128
data_selector - second instance
Refill block and active_offset
PENDING
Refill storage through existing wires
Data = mem_block; tag = active_tag; valid = 1
refill_write_en generation PENDING
refill_byte_data 8
cache_controller response - PENDING
resp_data 8 bits; resp_valid; resp_hit